

PROGRAM ON C

//Q1) Write a program to find the sum of 10 array elements.

```
#include <stdio.h>
```

```
int main() {
```

```
    int arr[10],sum=0;
```

```
    for(int i=0;i<10;i++)
```

```
    {
```

```
        printf("Enter a value of array arr[%d]:",i);
```

```
        scanf("%d",&arr[i]);
```

```
        sum+=arr[i];
```

```
    }
```

```
    printf("sum=%d",sum);
```

```
    return 0;
```

```
}
```

//Output : Enter a value of array arr[0]:1

Enter a value of array arr[1]:2

Enter a value of array arr[2]:3

Enter a value of array arr[3]:4

Enter a value of array arr[4]:5

Enter a value of array arr[5]:6

Enter a value of array arr[6]:7

Enter a value of array arr[7]:8

Enter a value of array arr[8]:9

Enter a value of array arr[9]:10

sum=55

//Q2) Write a program to input and display 5 array elements.

```
#include <stdio.h>
```

```
int main() {
```

```
    int arr[5];
```

```
    for(int i=0;i<5;i++)
```

```
    {
```

```
        printf("Enter a value of array arr[%d]:",i);
```

```
        scanf("%d",&arr[i]);
```

```
    }
```

```
    printf("array elements are");
```

```
    for(int i=0;i<5;i++)
```

```
    {
```

```
        printf("%d\t",arr[i]);
```

```
    }
```

```
    return 0;
```

```
}
```

//Output : Enter a value of array arr[0]:11

Enter a value of array arr[1]:22

Enter a value of array arr[2]:33

Enter a value of array arr[3]:44

Enter a value of array arr[4]:55

array elements are11 22 33 44 55

//Q3) Write a program to count even and odd numbers in an array.

```
#include <stdio.h>

int main() {

    int arr[10],even=0,odd=0;

    for(int i=0;i<10;i++)

    {

        printf("Enter a value of array arr[%d]:",i);

        scanf("%d",&arr[i]);

        if(arr[i]%2==0)

        {

            even++;

        }

        else

        {

            odd++;

        }

    }

    printf("Even=%d and odd=%d",even,odd);

    return 0;

}
```

//Output : Enter a value of array arr[0]:11

Enter a value of array arr[1]:22

Enter a value of array arr[2]:33

Enter a value of array arr[3]:44

Enter a value of array arr[4]:55

Enter a value of array arr[5]:66

Enter a value of array arr[6]:77

Enter a value of array arr[7]:88

Enter a value of array arr[8]:99

Enter a value of array arr[9]:11

Even=4 and odd=6

//Q4) Write a program to find the smallest and largest element in an array.

```
#include <stdio.h>
```

```
int main() {
```

```
    int arr[10]={2,3,1,11,4,10,33,7,2,5};
```

```
    int small,large,i;
```

```
    small=large=arr[0];
```

```
    for(int i=0;i<10;i++)
```

```
    {
```

```
    if(arr[i]<small)
```

```
    {
```

```
        small=arr[i];
```

```
    }
```

```
    if(arr[i]>large)
```

```
    {
```

```
        large=arr[i];
```

```
    }
```

```
    }
```

```
    printf("Small=%d and Large=%d",small,large);
```

```
    return 0;
```

```
}
```

```
//Output :Small=1 and Large=33
```

//Q5) Write a program to display all elements of a 4×3 matrix.

```
#include <stdio.h>

int main() {
    int matrix[4][3]={ {1,2,3},
        {4,5,6},
        {7,8,9},
        {10,11,12}
    };
    for(int i=0;i<4;i++)
    {
        for(int j=0;j<3;j++)
        {
            printf("matrix(%d)(%d)",i,j);
            printf("%d\t",matrix[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

//Q6)What is the output

```
#include <stdio.h>

#define FALSE -1
#define NULL 0
#define TRUE 1

int main() {
    if(NULL)
    {
        printf("NULL");
    }
}
```

```
    else if(FALSE)
    {
        printf("TRUE");
    }
    else
    {
        printf("FALSE");
    }
    return 0;
}
//Output : True
```

//Q7)

```
#include <stdio.h>

int main()
{
    int i=0,j=0;
    if(i++==j++)
    {
        printf("%d %d",i--,j--);
    }
    else
    {
        printf("%d %d",i,j);
    }
    return 0;
}

//Output 1 1
```

//Q8) Write a program to compare two strings.

```
#include <stdio.h>

int main()
{
    char str[8]="if block";
    if(str=="if block")
    {
        printf("if block id executed");
    }
    else
    {
        printf(" else block executed");
    }
    return 0;
}
```

//Output: else block executed

//Q9) Write a program to demonstrate the return value of printf().

```
#include <stdio.h>

int main()
{
    char str[]="\0";
    if(printf("%s",str))
    {
        printf("inside if block");
    }
    else
    {
        printf("inside else block");
    }
}
```

```
}  
    return 0;  
}
```

//Output: inside else block

//Q10)

```
#include <stdio.h>  
  
#define NULL 0  
  
int main()  
{  
    if (printf("0") == NULL)  
    {  
        printf("inside if block");  
    }  
    else  
    {  
        printf("inside else block");  
    }  
  
    return 0;  
}
```

//Q11) Write a program to demonstrate pre and post increment/decrement operators.

```
#include <stdio.h>  
  
int main()  
{  
    int i=5,j=4;  
    if(!printf(""))  
    {  
        printf("%d %d",i,j);  
    }  
}
```



```
}  
else  
{  
    printf("%d %d",i++,j++);  
}  
return 0;  
}  
//Output : 5 4
```

//Q12)

```
#include <stdio.h>  
  
int main()  
{  
    int i=1,j=0;  
    if(i--==j)  
    {  
        printf("i=%d",--i);  
    }  
    else  
    {  
        printf("j=%d",++j);  
    }  
    return 0;  
}
```

//Output : j= 1

//Q13)

```
#include <stdio.h>  
  
int main()  
{
```

```
int i = 25;
if(i == 25)
{
    i = 50;
}
if(i == 25)
{
    i = i + 1;
}
else
{
    i = i + 1;
}
printf("%d", i);
return 0;
}
//Output : 51
```

//Q14)

```
#include <stdio.h>
int main()
{
    int i = 2;
    if(i==(1,2))
    {
        printf("Hello ashish");
    }
    else
    {
        printf("Hello prashant");
    }
}
```

```
}  
    return 0;  
}
```

//Q15) Write a program to demonstrate the use of the sizeof() operator.

```
#include <stdio.h>
```

```
int main()
```

```
{  
    if(sizeof(0))  
    {  
        printf("Hi");  
    }  
    else  
    {  
        printf("Hello");  
    }  
    return 0;  
}
```

```
//O/p  Hi
```

```
// sizeof() will return size of integer so condition is true
```

//Q16) Write a program to find the size of a string literal using sizeof().

```
#include <stdio.h>
```

```
int main()
```

```
{  
    if(sizeof("\0"))  
    {  
        printf("inside if block");  
    }  
    else
```

```
{  
    printf("inside else");  
}  
return 0;  
}
```

//o/p inside if block

// sizeof("\0") is 1 byte returns true 1

//Q17)

```
#include <stdio.h>  
  
int main()  
{  
    int i,j;  
    for(int i=1,j=1;i<=3,j<=3;i++,j++)  
        printf("%d %d",i,j);  
    return 0;  
}  
  
//o/p : 112233
```

//Q18) Write a program to print numbers using a for loop.

```
#include <stdio.h>  
  
int main() {  
    int i;  
    for( i=0;i<=3;i++)  
    {  
        printf("%d",i);  
    }  
    return 0;  
}
```

//o/p 4

//Q19) Write a program to demonstrate an empty-body for loop.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i,j;
```

```
    for(;;)
```

```
        printf("%d ",20);
```

```
    return 0;
```

//Q19)

```
#include <stdio.h>
```

```
int main() {
```

```
    int i;
```

```
    for( i=0;i==0;i++);
```

```
    {
```

```
        printf("%d",i);
```

```
    }
```

```
    return 0;
```

```
}
```

//Q20) Write a program to demonstrate a while loop with post-increment.

```
#include <stdio.h>
```

```
int main() {
```

```
    int i=0;
```

```
    while(i++)
```

```
    {
```

```
        printf("loop");
```

```
        if(i==3)
        {
            break;
        }
    }
    return 0;
}
```

//Q21)

```
#include <stdio.h>

int main() {
    while(!printf("helpcode.com"));

    return 0;
}
```

//Output : helpcode.com

//Q22) Write a program to access array elements using alternative indexing notation.

```
#include <stdio.h>

int main() {
    int arr[1]={2};
    printf("%d",0[arr]);

    return 0;
}
```

//O/p 2

//Q23) Write a program to initialize and display array elements using loops.

```
#include <stdio.h>

int main() {
    int arr[3],i=0;
    for(int i=0;i<3;i++)
    {
        arr[i]=i;
    }
    for(int i=0;i<3;i++)
    {
        printf("%d",arr[i]);
    }
    return 0;
}

//o/p 012
```

//Q24) Write a program to access the last element of an array

```
#include <stdio.h>

int main() {
    int arr[3],i,j;
    arr[0]=11;
    arr[1]=22;
    arr[2]=33;

    for(int i=0;i<3;i++)
    {
        j=i;
    }
    printf("%d",arr[j]);
    return 0;
}
```

//Output 33

//Q25) Write a program to swap the first and last elements of an array.

```
#include <stdio.h>

int main() {
    int arr[3],i,temp;
    arr[0]=11;
    arr[1]=22;
    arr[2]=33;
    temp=arr[0];
    arr[0]=arr[2];
    arr[2]=temp;
    for(int i=0;i<3;i++)
    {
        printf("%d",arr[i]);
    }
    return 0;
}
```

//Output 332211

//Q26) Write a program to print a string character by character.

```
#include <stdio.h>

int main() {
    char name[]="Isha";
    int i;
    for(int i=0;i<6;i++)
    {
        printf("%c",name[i]);
    }
    return 0;
}
```



```
}  
//Output Isha
```

//Q27)

//Write a program to demonstrate different methods of array indexing.

```
#include <stdio.h>  
  
int main() {  
    char name[]="abcdef";  
    int i=0;  
        printf("%c",name[i]);  
        printf("%c",i[name]);  
        printf("%c",name[i+2]);  
        printf("%c",i[name+2]);  
    return 0;  
}  
//output aacc
```

//Q28) Write a program to access characters from a string defined

```
#include <stdio.h>  
  
#define arr "abcd"  
  
int main() {  
    printf("%c",arr[2]);  
  
    return 0;  
}  
//Output c
```

//Q29)

//Write a program to demonstrate array boundary conditions.

```
#include <stdio.h>
```

```
int main() {
```

```
    int arr[2]={1,2,3,4,5};
```

```
    printf("%d",arr[3]);
```

```
    return 0;
```

```
}
```

//Output 0 (in some other compiler)

//or

//Garbage value

//Q30)What will be the output

```
#include <stdio.h>
```

```
int main() {
```

```
    int i=0;
```

```
    char s[4]={'\0','\0','\0','\0'};
```

```
    for( i=0;i<4;i++)
```

```
    {
```

```
        printf("%c",s[i]);
```

```
    }
```

```
    return 0;
```

```
}
```

//o/p No output

//Q31)

//Write a program to demonstrate function calls with parameters and return values.

```
#include <stdio.h>

int function(int,int);

int main()
{
    int a=25,b=24+1,c;
    printf("%d",function(a,b));
    return 0;
}

int function(int x,int y)
{
    return(x-(x==y));
}

//Output 24
```

//Q32) Write a program to demonstrate the effect of post-increment in function

```
#include <stdio.h>

int _a_1234(int n);

int main()
{
    int n = _a_1234(4);
    printf("%d\n", --n);
    return 0;
}

int _a_1234(int n)
{
```

```
    return (n++);  
}  
//Output 3
```

//Q33)

//Write a program to demonstrate pointer arithmetic on character arrays.

```
#include <stdio.h>  
char abc(char arr[]);  
int main()  
{  
    char arr[100];  
    arr[0] = 'a';  
    arr[1] = 'b';  
    arr[2] = 'c';  
    arr[3] = 'd';  
    abc(arr);  
    return 0;  
}  
char abc(char arr[])  
{  
    printf("%c", *++arr);  
    printf("%c", *arr++);  
}  
//Output bb
```

//Q34)

//Write a program to demonstrate type conversion .

```
#include <stdio.h>

int show(int returns);

int main()
{
    int num = show(sizeof(float));
    printf("Value is %d", ++num);
    return 0;
}

int show(int returns)
{
    returns += 5.31;
    return returns;
}

//Output Value is 10
```

//Q35)

//Write a program to demonstrate the use of static local variables.

```
#include <stdio.h>

int num = 5;

int fun();

int main()
{
    fun();
    fun();
    return 0;
}
```

```

int fun()
{
    static int num = 2;
    printf("%d", num);
    num++;
    return 0;
}
//output 2 3

```

//Q36) Write a program to return a dynamically allocated integer from a function.

```

#include <stdio.h>
#include <stdlib.h>
int *fun();
int main()
{
    int *ptr = fun();
    printf("%d", *ptr);
    free(ptr);
    return 0;
}
int *fun()
{
    int *ptr1 = (int *)malloc(sizeof(int));
    *ptr1 = 25;
    return ptr1;
}
//Output 25

```

//Q37) Write a program to demonstrate recursion.

```
#include <stdio.h>
```

```
void fun(int);
```

```
int main()
```

```
{
```

```
    int a=3;
```

```
    fun(a);
```

```
    return 0;
```

```
}
```

```
void fun(int n)
```

```
{
```

```
    if(n>0)
```

```
    {
```

```
        fun(--n);
```

```
        printf("%d",n);
```

```
    }
```

```
}
```

```
//Output 012
```

//Q38) Program on printing the addresses of integer and float variables.

```
#include <stdio.h>
```

```
int main() {
```

```
    int age=30;
```

```
    float sal=1500.50;
```

```
    printf("Address of age=%u\n",&age);
```

```
    printf("Address of salary=%u\n",&sal);
```

```
    return 0;
```

```
}
```

```
//Output: Address of age=2442743660
```

```
//Address of salary=2442743656
```

//Q39)Program for dereferencing pointer

```
#include <stdio.h>

int main() {

    int a=87;

    int *p1=&a;

    printf("Address of a=%u\n ",&a);

    printf("Address of p1=%u\n ",&p1);

    printf("Value of p1=%d\n ",&p1);

    printf("Value of a=%d\n ",&a);

    printf("Value of a=%d\n ",&p1);

    printf("Value of a=%d\n ",*(&a));

    return 0;

}
```

//Output

Address of a=4261745468

Address of p1=4261745456

Value of p1=-33221840

Value of a=-33221828

Value of a=-33221840

Value of a=87

//Q40) Program on pointer to pointer (double pointer).

```
#include <stdio.h>
```

```
int main()
```

```
{
```



```

int a = 87;

int *p1 = &a;

int **q1 = &p1;

printf("Address of a = %u\n", &a);
printf("Address of p1 = %u\n", &p1);
printf("Address of q1 = %u\n", &q1);
printf("Value of p1 %d\n", p1);
printf("Value of q1 %d\n", q1);
printf("Value of a %d\n", a);
printf("Value of a %d\n", *p1);
printf("Value of a %d\n", *(&a));
printf("Value of a %d\n", **q1);

return 0;
}

```

//Output

```

Address of a = 432396460
Address of p1 = 432396448
Address of q1 = 432396440
Value of p1 432396460
Value of q1 432396448
Value of a 87
Value of a 87
Value of a 87
Value of a 87

```

//Q41) Print the size of pointer and size of value

//In C

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```

char a = 'x', *p1 = &a;
int b = 12, *p2 = &b;
float c = 2.5f, *p3 = &c;
double d = 18.33, *p4 = &d;
printf("sizeof(p1) = %zu, sizeof(*p1) = %zu\n", sizeof(p1), sizeof(*p1));
printf("sizeof(p2) = %zu, sizeof(*p2) = %zu\n", sizeof(p2), sizeof(*p2));
printf("sizeof(p3) = %zu, sizeof(*p3) = %zu\n", sizeof(p3), sizeof(*p3));
printf("sizeof(p4) = %zu, sizeof(*p4) = %zu\n", sizeof(p4), sizeof(*p4));
return 0;
}

```

//Output

```

sizeof(p1) = 8, sizeof(*p1) = 1
sizeof(p2) = 8, sizeof(*p2) = 4
sizeof(p3) = 8, sizeof(*p3) = 4
sizeof(p4) = 8, sizeof(*p4) = 8

```

//Q42) Program on pointer arithmetic.

```

#include <stdio.h>

int main()
{
    int a = 5, *pi = &a;
    char b = 'x', *pc = &b;
    float c = 5.5, *pf = &c;
    printf("value of pi=Address of a=%u\n",pi);
    printf("value of pc=Address of b=%u\n",pc);
    printf("value of pf=Address of c=%u\n",pf);
    pi++;
    pc++;
    pf++;
    printf("now value of pi=%u\n",pi);
    printf("now value of pc=%u\n",pc);
}

```

```

printf("now value of pf=%u\n",pf);
return 0;
}
//Output
value of pi=Address of a=1727614740
value of pc=Address of b=1727614739
value of pf=Address of c=1727614732
now value of pi=1727614744
now value of pc=1727614740
now value of pf=1727614736

```

//Q43) Program on accessing array elements using different pointer notations.

```

#include <stdio.h>
int main()
{
    int arr[5]={5,10,20,25,30},i=0;
    for(i=0;i<5;i++)
    {
        printf("value of a[%d]\n",i);
        printf("%d\n",arr[i]);
        printf("%d\n",i[arr]);
        printf("%d\n",*(arr+i));
        printf("%d\n",*(i+arr));
    }
    return 0;
}

```

//Output: value of a[0]

5
5
5

5

value of a[1]

10

10

10

10

value of a[2]

20

20

20

20

value of a[3]

25

25

25

25

value of a[4]

30

30

30

30

//Q44) Program on typecasting an integer pointer to a character pointer.

//In C

#include <stdio.h>

int main()

{

int a=66;

char*c;

c=(char*)&a;

printf("%d\n",*c);

```
    printf("%c\n", *c);  
    return 0;  
}
```

//Output:66

B

//Q45) Program on passing an array to a function using pointers.

```
#include <stdio.h>  
  
void fun(int *p);  
  
int main()  
{  
    int arr[] = {3, 6, 9, 12, 15, 18};  
    fun(arr);  
    return 0;  
}  
  
void fun(int *p)  
{  
    int i;  
    for(i = 0; i < 6; i++)  
    {  
        printf("%d\n", *p);  
        p++;  
    }  
}
```

//Output

3

6

9

12

15

18

//Q46) Program on NULL pointers.

```
#include <stdio.h>

int main()
{
    int *ptr=NULL;
    printf("The Value of ptr is %p",ptr);
    return 0;
}
```

//Output:

//The Value of ptr is (nil)

//Q47) Program on reading a string using getchar().

```
#include <stdio.h>

int main() {
    char name[10],ch;
    int i=0;
    printf("Entre name");
    while(i<10)
    {
        ch=getchar();
        name[i]=ch;
        i++;
    }
    name[i]='\0';
    printf("Name:%s",name);
    return 0;
}
```

//Q48) Program on reading a string until the Enter key is pressed.

```
#include <stdio.h>
```

```
int main()
{
    char name[10], ch;
    int i = 0;
    printf("Enter name: ");
    while (i < 10)
    {
        ch = getchar();
        if (ch == '\n')
            break;
        name[i] = ch;
        i++;
    }
    name[i] = '\0';
    printf("Name: %s", name);
    return 0;
}

//Output:
//Enter name: isha
//Name: isha
```

//Q49) Program on comparing two strings using strcmp().

```
#include<string.h>
#include <stdio.h>
int main()
{
    char str1[150],str2[50];
    printf("Enter the first string\n ");
    scanf("%s",str1);
    printf("Enter the second string\n ");
    scanf("%s",str2);
```

```

if(strcmp(str1,str2)==0)
{
    printf("Equal\n");
}
else
{
    printf("Not equal\n");
}
return 0;
}

```

//Output

//Enter the first string

//isha

//Enter the second string

//isha

//Equal

//Q50) Program on finding the length of strings using strlen().

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char a[20] = "Isha";
```

```
    char b[20] = {'I', 'S', 'H', 'A', '\0'};
```

```
    char c[20];
```

```
    printf("Enter string: ");
```

```
    scanf("%s", c);
```

```
    printf("Length of string a = %d\n", strlen(a));
```

```
    printf("Length of string b = %d\n", strlen(b));
```

```
    printf("Length of string c = %d\n", strlen(c));
```



```
    return 0;
}

//Output: Enter string: help4code
Length of string a = 4
Length of string b = 4
Length of string c = 9
```

//Q51) Program on validating login credentials using string lengths.

```
#include <stdio.h>
#include <string.h>

int main()
{
    char username[60];
    char password[90];
    printf("Enter username: ");
    scanf("%s", username);
    printf("Enter password: ");
    scanf("%s", password);
    int userLength = strlen(username);
    int passLength = strlen(password);
    printf("Length of Username: %d\n", userLength);
    printf("Length of Password: %d\n", passLength);
    if(userLength == passLength)
    {
        printf("Login Success\n");
    }
    else
    {
        printf("Login Failed\n");
    }
    return 0;
}
```

//Output

Enter username: Isha

Enter password: 1234

Length of Username: 4

Length of Password: 4

Login Success

//Q52) Program on finding the length of a string using a user-defined function.

```
#include <stdio.h>
```

```
void mystrlen(char *);
```

```
int main()
```

```
{
```

```
    char str2[20];
```

```
    printf("Enter any string: ");
```

```
    scanf("%s", str2);
```

```
    mystrlen(str2);
```

```
    return 0;
```

```
}
```

```
void mystrlen(char *s1)
```

```
{
```

```
    int length = 0;
```

```
    while (*s1 != '\0')
```

```
    {
```

```
        length++;
```

```
        s1++;
```

```
    }
```

```
    printf("Length is %d", length);
```

```
}
```

//Output

Enter any string: help4code

Length is 9

//Q53) Program on converting an uppercase string to lowercase.

```
#include <stdio.h>

void mystrlwr(char *s);

int main()
{
    char ch[20];
    printf("Enter any string: ");
    scanf("%s", ch);
    mystrlwr(ch);
    printf("\n Upper case String");
    scanf("%s", ch);
    return 0;
}

void mystrlwr(char *s)
{
    while (*s != '\0')
    {
        *s=*s+32;
        s++;
    }
}
```

//Output:

Enter any string: ISHA

Upper case String isha

//Q54) Program on counting the number of words in a string.

```
#include <stdio.h>

int main()
{
    char s[200];
    int count = 0, i;

    printf("Enter the string: ");
    scanf("%[^\n]", s);

    for(i = 0; s[i] != '\0'; i++)
    {
        if(s[i] == ' ')
            count++;
    }

    printf("Number of words in given string are: %d", count + 1);
    return 0;
}
```

//Output:

Enter the string: c is program

Number of words in given string are: 3

//Q55) Program on switch-case

```
#include <stdio.h>

int main()
```

```
{  
    int i = 1;  
  
    switch(i)  
    {  
        case 1:  
            printf("HI");  
  
        default:  
            printf("HELLO");  
    }  
    return 0;  
}  
//Output  
//HI HELLO
```

//Q56) Program on switch-case using character values.

```
#include <stdio.h>  
  
int main()  
{  
    char ch = 65;  
    switch(ch)  
    {  
        case 'A':  
            printf("apple");  
            break;  
        case 'B':  
            printf("Bing");  
            break;  
        default:
```

```
        printf("Bye");
    }
    return 0;
}
```

//Output: apple

//Q57) Program on switch-case with integer and character constants having the same value.

```
#include <stdio.h>

int main()
{
    int i = 65;
    switch(i)
    {
        case 65:
            printf("interger 65");
            break;
        case 'A':
            printf("Char 65");
            break;
    }
    return 0;
}
```

//Q58) Program on demonstrating the comma operator in a switch expression.

```
#include <stdio.h>

int main()
{
    int i = 65;
    char ch='B';
```

```
switch(ch,i)
{
    case 65:
        printf("integer");
        break;
    case 'B':
        printf("Char");
        break;
}
return 0;
}
```

//Output: integer

//Q59) Program on demonstrating increment and decrement operators

```
#include <stdio.h>
int main()
{
    int i = 1;
    i++;
    switch(i--)
    {
        case 1:
            printf("case1");
            break;
        case 2:
            printf("case2");
            break;
        default:
            printf("default");
    }
    return 0;
}
```

```
}  
//Output: case2
```

//Q60) Program on demonstrating switch-case execution without a break statement.

```
#include <stdio.h>  
int main()  
{  
    switch(3/2)  
    {  
        case 1:  
            printf("case1");  
        case 2:  
            printf("case2");  
            break;  
        default:  
            printf("default");  
    }  
    return 0;  
}  
//Output: case1case2
```

//Q61) Program on demonstrating the comma operator in an if condition and switch statement.

```
#include <stdio.h>  
int main()  
{  
    int num=0;
```



```
if(num++,num--,++num)
switch(num)
{
    case 1:
        printf("case1");
        break;
    case 2:
        printf("case2");
        break;
    default:
        printf("default");
        break;
}
return 0;
}
//Output: case1
```

//Q62)What will be the output

```
#include <stdio.h>
int main()
{
    int num = 5;
    switch(num++ == 5)
    {
        case 1:
            printf("true");
            break;
        case 2:
            printf("false");
            break;
```

```
    default:
        printf("default");
    }
    return 0;
}
//Output: true
```